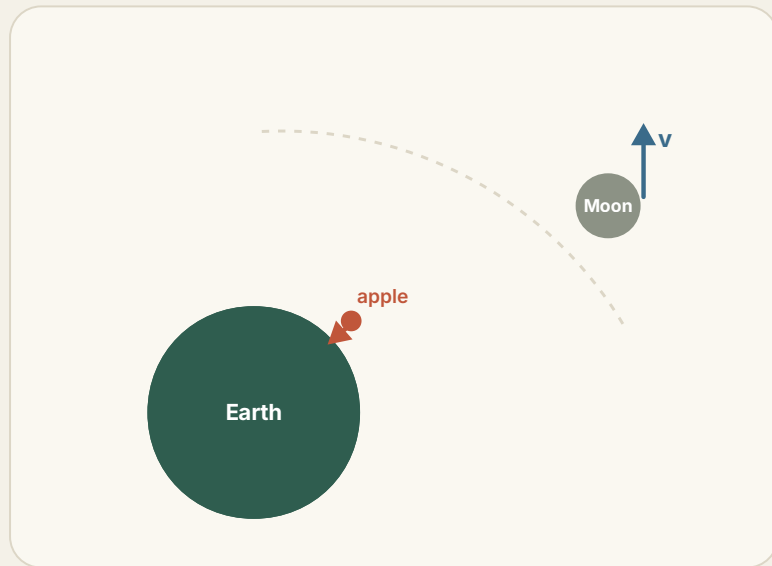


Universal Gravitation

The same pull that drops an apple keeps the Moon swinging around the Earth.

ESSENTIAL QUESTION

Why should one rule govern both a falling apple and the orbiting Moon — and how strong is that pull?



● LESSON MAP

Build the law in four moves.

01

GUESS

Maybe the apple and the Moon feel the very same pull.

02

TEST

Does the pull weaken with distance? Check the Moon's fall against g .

03

LAW

Force grows with both masses and dies away as $1/r^2$.

04

USE

Measure G , then weigh planets and predict orbits.

KEEP THIS IN MIND

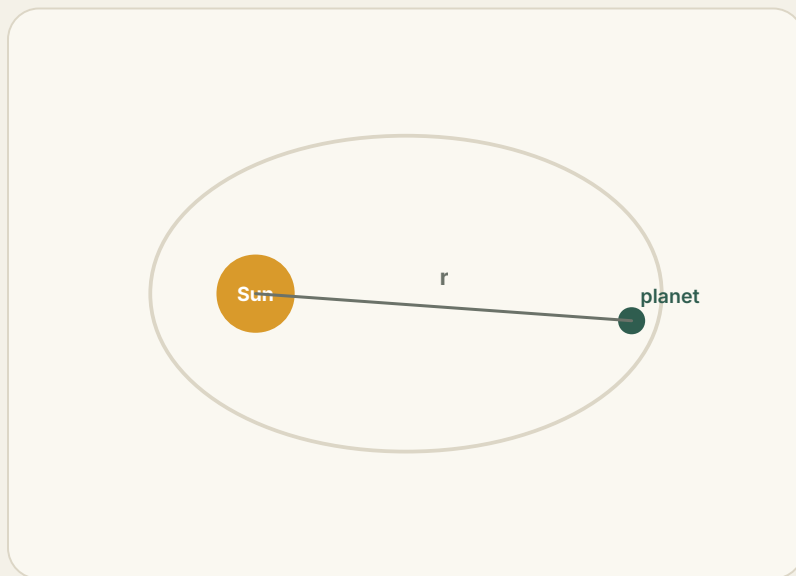
Gravity is **universal**, **mutual**, and fades as the **square** of the distance.

RECALL

Kepler mapped the orbits — not the cause.

By 1619 the planets' motion was fully described. Yet nothing said why they keep turning.

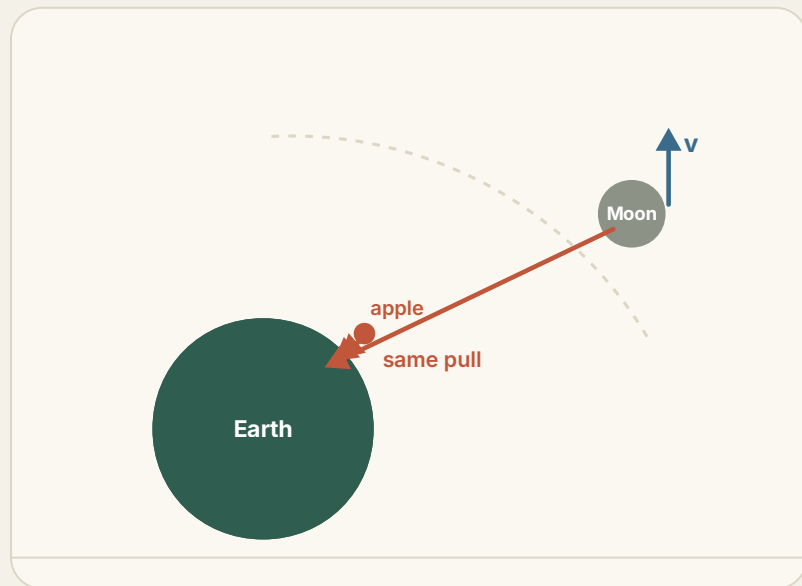
- 1 Orbits are ellipses**
With the Sun at one focus.
- 2 Equal areas in equal times**
A planet speeds up when it is closer.
- 3 A period–size rule**
Ties the year to the orbit.



THE BIG IDEA

One force reaches from the ground to the sky.

Newton's leap: the pull that bends the apple's path is the same pull that bends the Moon's.



The Moon is really **falling** toward Earth all the time — it just moves sideways fast enough to keep missing.

If one force does both jobs, it should weaken in a way we can measure. Let's check.

THE TEST

How fast does the Moon really fall?

The Moon sits about 60 Earth-radii away. Newton compared its fall to a stone's.

1 Sixty radii out

The Moon's distance is about 60 R from Earth's centre.

2 The pull is 3600× weaker

Its tiny fall, measured against surface gravity g.

3 And 3600 = 60²

No accident — the force falls off as the square.

THE VERDICT

Sixty times farther, and the pull is 3600 times weaker.

SO THE FORCE MUST OBEY

$$F \propto \frac{1}{r^2}$$

REASONING

What the force has to depend on.

Three clues fix its shape — then one constant glues them together.

1 Both masses

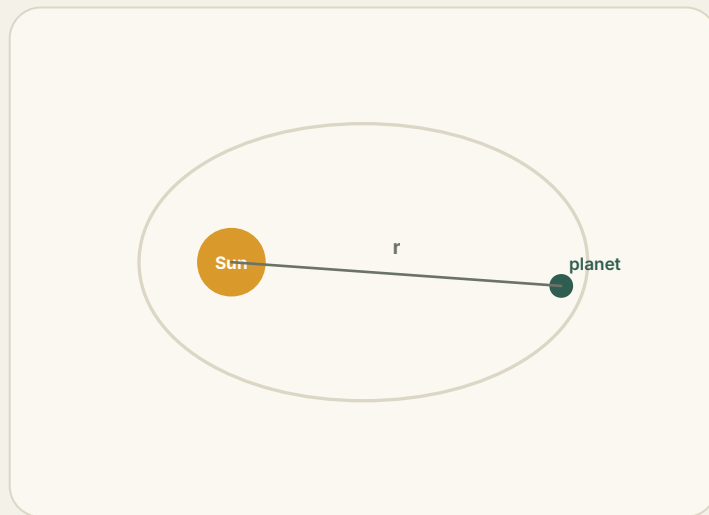
The pull is mutual, so it grows with each body: $F \propto m_1$,
 $F \propto m_2$.

2 The distance

The Moon test fixes the fall-off: $F \propto \frac{1}{r^2}$.

3 One constant to glue it

Fold the proportionality into a single number, G .



THE LAW

Newton's Law of Universal Gravitation.

$$F = G \frac{m_1 m_2}{r^2}$$

m_1, m_2 the two masses (kilograms)

r distance between their centres (metres)

G the gravitational constant

WHEN MAY I USE IT?

For two point masses, or uniform spheres — then r runs centre to centre. Close, oddly-shaped bodies need calculus.

CHARACTER

Four promises the law makes.

01 · Universal

Every pair of masses attracts — apples, planets, galaxies, you.

03 · Inverse-square

Double the distance and the pull drops to a quarter.

02 · Mutual

Equal and opposite, by Newton's third law: $F_{12} = F_{21}$.

04 · Along the line

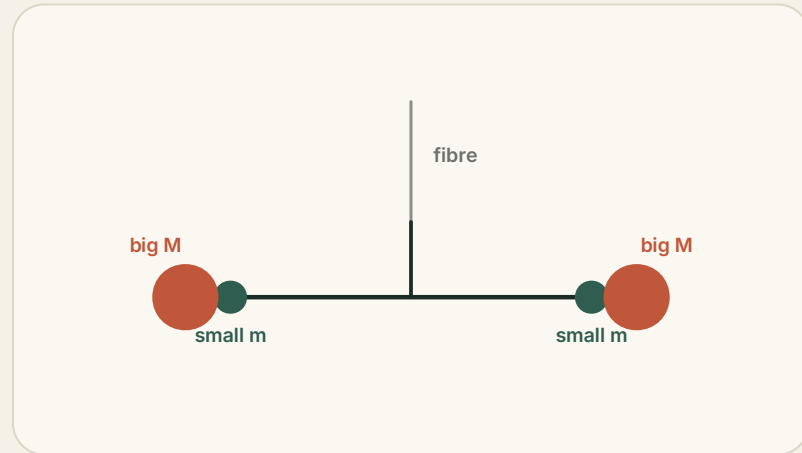
It points straight from each mass toward the other — always attractive.

ONE PICTURE TO HOLD $r \rightarrow 2r \Rightarrow F \rightarrow \frac{1}{4}F$

THE CONSTANT

Cavendish weighed the world.

A delicate torsion balance let him read the tiny pull between lab masses — and so fix G .



So small that gravity is the **weakest** of the forces — only huge masses make it felt.

ONCE G IS KNOWN, WEIGH THE EARTH

$$M = \frac{gR^2}{G}$$

WORKED EXAMPLE

Why you don't feel the person beside you.

Two students, 60 kg each, standing 1 m apart. How hard do they pull?

1 Drop the numbers in

$$F = \frac{(6.67 \times 10^{-11})(60)(60)}{1^2}$$

2 So the mutual pull is

$$F \approx 2.4 \times 10^{-7} \text{ N} \text{ — a speck of dust.}$$

3 Earth, meanwhile, pulls with

$$mg = 60 \times 9.8 \approx 590 \text{ N}$$

THE LESSON

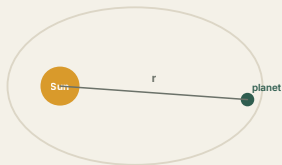
The law always applies — but it takes a planet-sized mass before you ever notice the pull.

● SUMMARY

Every mass pulls every other — universal, mutual, fading as $1/r^2$.

NEWTON'S LAW OF UNIVERSAL GRAVITATION

$$F = G \frac{m_1 m_2}{r^2}$$



Pin down G , and the whole solar system falls into place.

Distance matters far more than mass — double r and the pull drops to a quarter.